

NAT & Private IPv4 Addresses

1. IPv4 Address Shortage

IPv4 provides about **4.3 billion addresses**, which is not enough for every device on the Internet.

The long-term solution is **IPv6**.

Before IPv6 became widely adopted, several techniques helped deal with IPv4 address exhaustion:

1. **CIDR** — Classless Inter-Domain Routing
2. **Private IPv4 Addresses**
3. **NAT** — Network Address Translation

Why Private IP + NAT?

Private IP addresses allow organizations to use the same IP ranges internally.

NAT then allows those private IP addresses to communicate with the Internet using public IP addresses.

IMAGE PLACEMENT 1



Private IPv4 Addresses (RFC 1918)

- IPv4 doesn't provide enough addresses for all devices that need an IP address in the modern world.
- The long-term solution is to switch to IPv6.
- There are three main short-term solutions:
 - 1) CIDR
 - 2) Private IPv4 addresses
 - 3) NAT
- RFC 1918 specifies the following IPv4 address ranges as private:

10.0.0.0/8 (10.0.0.0 to 10.255.255.255)  Class A

172.16.0.0/12 (172.16.0.0 to 172.31.255.255)  Class B

192.168.0.0/16 (192.168.0.0 to 192.168.255.255)  Class C



2. Private IPv4 Addresses

Private IPv4 addresses are defined by **RFC 1918**.

There are three private IPv4 ranges:

Private Range	CIDR	Address Range
10.0.0.0/8	/8	10.0.0.0 – 10.255.255.255
172.16.0.0/ 12	/12	172.16.0.0 – 172.31.255.255
192.168.0.0 /16	/16	192.168.0.0 – 192.168.255.255

Important

These addresses:

- Can be freely used inside private networks.
- Do **not** need to be globally unique.
- Are **not routable over the public Internet**.

For example:

PC1 = 192.168.0.167

PC2 = 192.168.0.168

Two different organizations can use the same addresses internally.

Why?

Because these addresses are intended for **private/local networks**, not direct Internet communication.

Two Important Problems

1. Duplicate addresses

Different private networks can use the same IP addresses.

Company A → 192.168.0.167

Company B → 192.168.0.167

This is allowed because they are separate private networks.

2. Private IPs cannot be used directly over the Internet

A PC with:

192.168.0.167

cannot simply send an Internet packet using that private address as its Internet-routable source address.

This is where **NAT** becomes important.

📌 IMAGE PLACEMENT 2



Private IPv4 Addresses (RFC 1918)

- RFC 1918 specifies the following IPv4 address ranges as private:
 - 10.0.0.0/8 (10.0.0.0 to 10.255.255.255)
 - 172.16.0.0/12 (172.16.0.0 to 172.31.255.255)
 - 192.168.0.0/16 (192.168.0.0 to 192.168.255.255)
- You are free to use these addresses in your networks. They don't have to be globally unique.

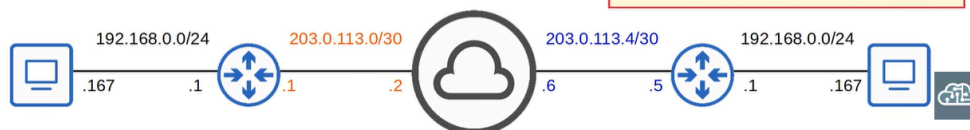
```
C:\Users\user>ipconfig
Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : 
    IPv4 Address. . . . . : 192.168.0.167
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.0.1
```

- Two problems:
 - 1) Duplicate addresses
 - 2) Private IP addresses can't be used over the Internet, so the PCs can't access the Internet.

*Private IP addresses cannot be used over the Internet!



3. Network Address Translation — NAT

NAT = Network Address Translation

NAT modifies the **source and/or destination IP address** of packets.

The most common use of NAT is:

Allowing devices with private IP addresses to communicate with hosts on the Internet.

Example:

Before NAT:

Source: 192.168.0.167

Destination: 8.8.8.8

R1 translates the source:

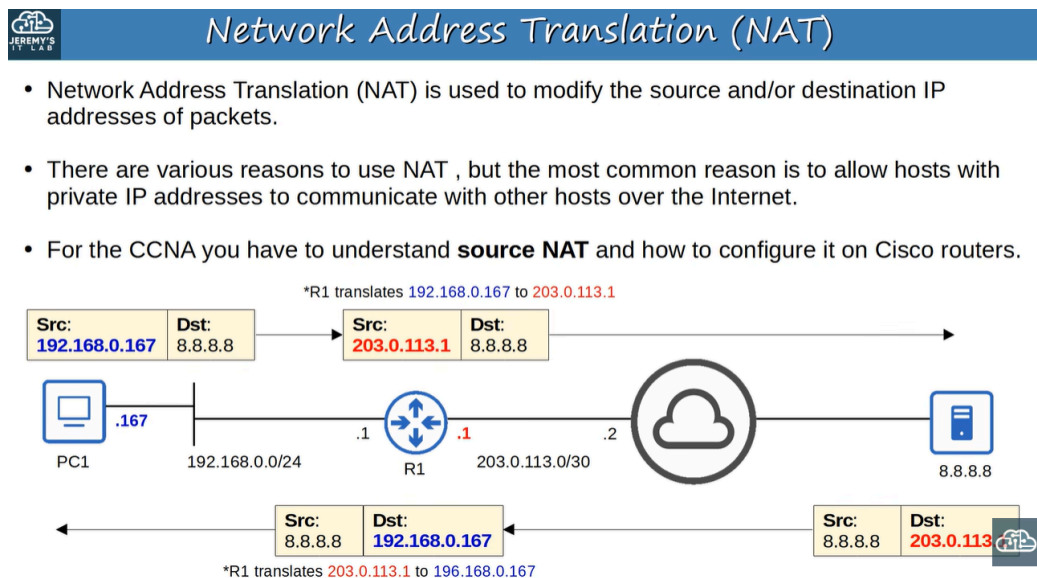
After NAT:

Source: 203.0.113.1

Destination: 8.8.8.8

The Internet therefore sees the **public address**, not the private address.

IMAGE PLACEMENT 3



4. Static NAT

Static NAT creates a permanent **one-to-one mapping** between a private IP address and a public IP address.

Example:

192.168.0.167 ↔ 100.0.0.1
192.168.0.168 ↔ 100.0.0.2

Therefore:

Private IP		Public IP
Inside Local	→	Inside Global

192.168.0.167	→	100.0.0.1
192.168.0.168	→	100.0.0.2

Important limitation

Static NAT requires **one public IP for each private IP**.

For example:

100 private devices
↓
100 public IP addresses required

Therefore, Static NAT does **not conserve public IPv4 addresses effectively**.

5. NAT Terminology

This is very important for CCNA.

Inside Local

The IP address of the inside host **from the perspective of the local/private network**.

Usually:

Private IP address configured on the inside device.

Example:

Inside Local = 192.168.0.167

Inside Global

The IP address of the inside host **from the perspective of the outside network**.

Usually:

Public IP address assigned to the inside host after NAT.

Example:

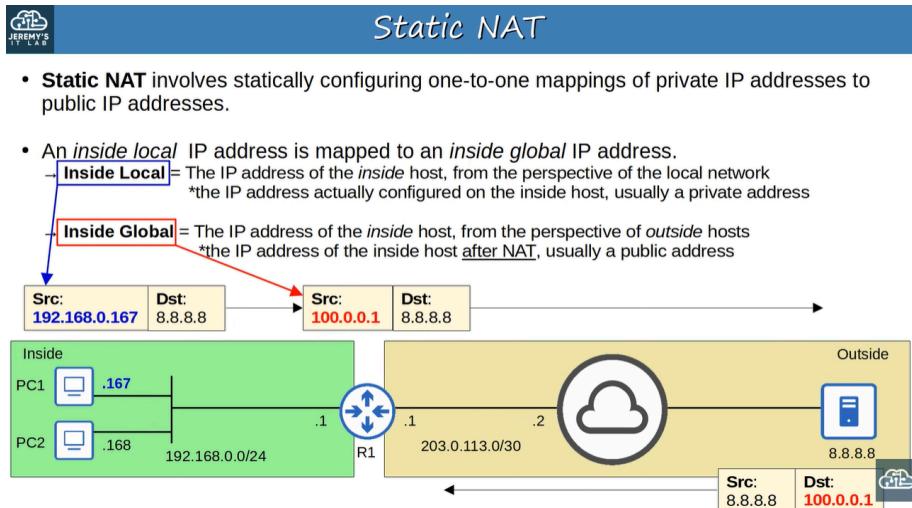
Inside Global = 100.0.0.1

Easy memory trick

Inside Local = Inside device's real/private IP

Inside Global = Inside device's public/NAT IP

IMAGE PLACEMENT 4



6. Static NAT — Example with PC2

Suppose:

PC2

Private IP = 192.168.0.168

Static NAT maps it to:

Public IP = 100.0.0.2

When PC2 sends:

Source: 192.168.0.168

Destination: 8.8.8.8

R1 changes it to:

Source: 100.0.0.2

Destination: 8.8.8.8

The reply from 8.8.8.8 is:

Source: 8.8.8.8

Destination: 100.0.0.2

R1 translates the destination back:

Source: 8.8.8.8

Destination: 192.168.0.168

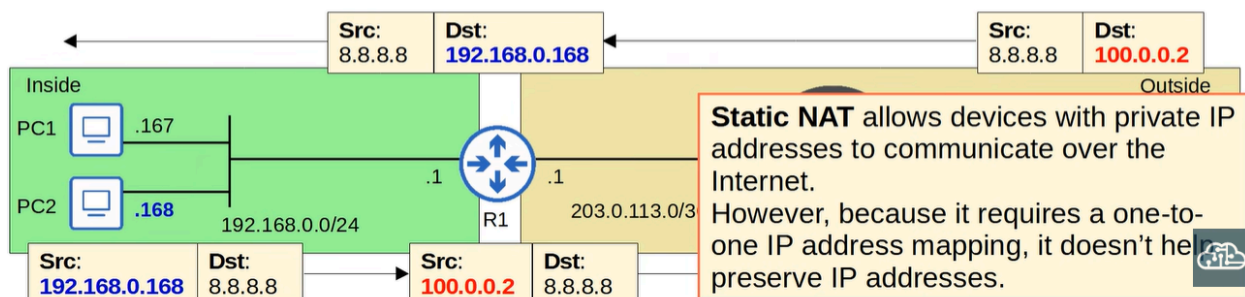
So PC2 receives the reply.

📌 IMAGE PLACEMENT 5



Static NAT

- **Static NAT** involves statically configuring one-to-one mappings of private IP addresses to public IP addresses.
- An *inside local* IP address is mapped to an *inside global* IP address.
 - **Inside Local** = The IP address of the *inside* host, from the perspective of the local network
*the IP address actually configured on the inside host, usually a private address
 - **Inside Global** = The IP address of the *inside* host, from the perspective of *outside* hosts
*the IP address of the inside host after NAT, usually a public address



7. Static NAT Configuration

First identify the interfaces.

Inside interface

The interface connected to the **private/internal network**:

```
R1(config)# interface g0/1
R1(config-if)# ip nat inside
```

Outside interface

The interface connected to the **external/Internet network**:

```
R1(config)# interface g0/0
R1(config-if)# ip nat outside
```

Create the Static NAT mapping

Syntax:

```
ip nat inside source static inside-local-ip inside-global-ip
```

Example:

```
R1(config)# ip nat inside source static 192.168.0.167 100.0.0.1
R1(config)# ip nat inside source static 192.168.0.168 100.0.0.2
```

Complete configuration

```
R1(config)# interface g0/1
R1(config-if)# ip nat inside
R1(config-if)# exit
```

```
R1(config)# interface g0/0
R1(config-if)# ip nat outside
R1(config-if)# exit
```

```
R1(config)# ip nat inside source static 192.168.0.167 100.0.0.1
R1(config)# ip nat inside source static 192.168.0.168 100.0.0.2
```

 **IMAGE PLACEMENT 6**

Static NAT Configuration

```

R1(config)#int g0/1
R1(config-if)#ip nat inside
R1(config-if)#int g0/0
R1(config-if)#ip nat outside
R1(config-if)#exit

R1(config)#ip nat inside source static 192.168.0.167 100.0.0.1
R1(config)#ip nat inside source static 192.168.0.168 100.0.0.2
R1(config)#exit

```

R1#show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
udp	100.0.0.1:56310	192.168.0.167:56310	8.8.8.8:53	8.8.8.8:53
---	100.0.0.1	192.168.0.167	---	---
udp	100.0.0.2:62321	192.168.0.168:62321	8.8.8.8:53	8.8.8.8:53
---	100.0.0.2	192.168.0.168	---	---

Inside

PC1 .167

PC2 .168

192.168.0.0/24

R1

Outside

203.0.113.0/30

8.8.8.8

8. show ip nat translations

Use:

R1# show ip nat translations

Example:

```

Pro Inside global Inside local
--- 100.0.0.1      192.168.0.167
--- 100.0.0.2      192.168.0.168

```

Meaning

Inside Global		Inside Local
100.0.0.1	↔	192.168.0.167
100.0.0.2	↔	192.168.0.168

The static mappings remain even when there is no active traffic.

When PC1 or PC2 communicates with 8.8.8.8, additional information can appear:

```

Inside Global
Inside Local
Outside Local
Outside Global

```

9. Outside Local and Outside Global

These terms describe the **outside host** from different perspectives.

Outside Local

The IP address of the outside host **from the perspective of the local network**.

Outside Global

The IP address of the outside host **from the perspective of the outside network**.

In normal source NAT:

Outside Local = Outside Global

unless destination NAT is being used.

Easy memory trick

Inside / Outside = Location of the host

Local / Global = Perspective

So:

Inside Local

Inside Global

Outside Local

Outside Global



IMAGE PLACEMENT 7



show ip nat translations

```
R1#show ip nat translations
Pro Inside global   Inside local   Outside local   Outside global
udp 100.0.0.1:56310 192.168.0.167:56310 8.8.8.8:53      8.8.8.8:53
--- 100.0.0.1       192.168.0.167   ---            ---
udp 100.0.0.2:62321 192.168.0.168:62321 8.8.8.8:53      8.8.8.8:53
--- 100.0.0.2       192.168.0.168   ---            ---
```

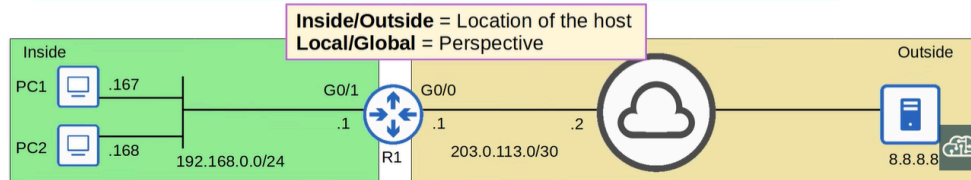
Unless **destination NAT** is used, these two addresses will be the same.

→ **Inside Local** = The IP address of the *inside* host, from the perspective of the local network
*the IP address actually configured on the inside host, usually a private address

→ **Inside Global** = The IP address of the *inside* host, from the perspective of *outside* hosts
*the IP address of the inside host after NAT, usually a public address

→ **Outside Local** = The IP address of the *outside* host, from the perspective of the local network

→ **Outside Global** = The IP address of the *outside* host, from the perspective of the outside network



10. Clearing NAT Translations

Command:

```
R1# clear ip nat translation *
```

The * means:

Clear all dynamic NAT translation entries.

After clearing, static mappings remain.

For example:

Before:

100.0.0.1 ↔ 192.168.0.167

100.0.0.2 ↔ 192.168.0.168

UDP active translations...

After:

100.0.0.1 ↔ 192.168.0.167

100.0.0.2 ↔ 192.168.0.168

The **static mappings remain** because they are configured permanently.

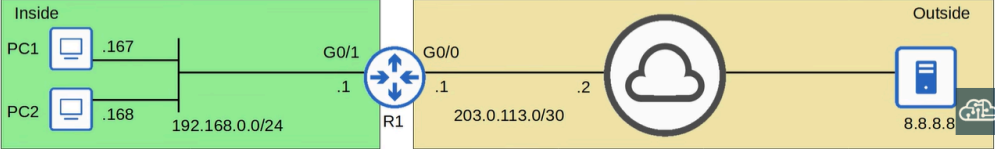
IMAGE PLACEMENT 8

clear ip nat translation *

```
R1#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
udp 100.0.0.1:56310    192.168.0.167:56310 8.8.8.8:53         8.8.8.8:53
--- 100.0.0.1          192.168.0.167      ---                ---
udp 100.0.0.2:62321    192.168.0.168:62321 8.8.8.8:53         8.8.8.8:53
--- 100.0.0.2          192.168.0.168      ---                ---

R1#clear ip nat translation *

R1#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
--- 100.0.0.1          192.168.0.167      ---                ---
--- 100.0.0.2          192.168.0.168      ---                ---
```



11. show ip nat statistics

Command:

```
R1# show ip nat statistics
```

This displays information about NAT operation.

Important information includes:

Total active translations

Inside interfaces

Outside interfaces

Hits

Misses

Dynamic mappings

Example:

Total active translations: 2

2 static

0 dynamic

It also shows:

Inside interfaces:
GigabitEthernet0/1

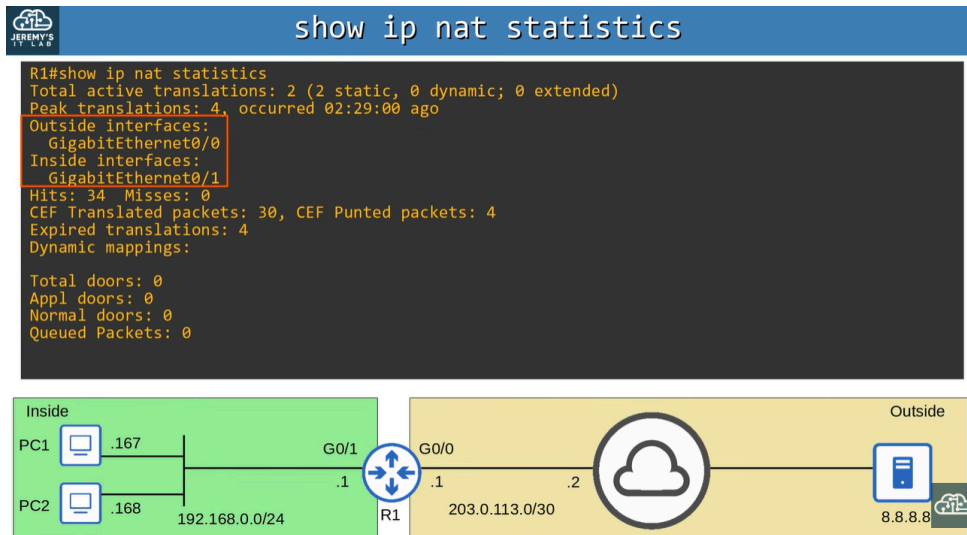
Outside interfaces:
GigabitEthernet0/0

Important

From the example:

G0/1 = Inside
G0/0 = Outside

📌 IMAGE PLACEMENT 9



12. Important NAT Commands

Configure inside interface

R1(config-if)# ip nat inside

Configure outside interface

R1(config-if)# ip nat outside

Configure Static NAT

R1(config)# ip nat inside source static inside-local-ip inside-global-ip

Example:

```
R1(config)# ip nat inside source static 192.168.0.167 100.0.0.1
```

Display NAT translations

```
R1# show ip nat translations
```

Display NAT statistics

```
R1# show ip nat statistics
```

Clear NAT translations

```
R1# clear ip nat translation *
```

IMAGE PLACEMENT 10

```
R1(config-if)# ip nat inside
R1(config-if)# ip nat outside
R1(config)# ip nat inside source static inside-local-ip inside-global-ip
R1# show ip nat translations
R1# show ip nat statistics
R1# clear ip nat translation *
```

13. NAT Packet Flow — Remember This

Suppose:

PC1 = 192.168.0.167

NAT = 100.0.0.1

Server = 8.8.8.8

Outgoing packet

PC1

192.168.0.167

↓

R1

↓

100.0.0.1

↓

8.8.8.8

Original:

Source = 192.168.0.167

Destination = 8.8.8.8

After NAT:

Source = 100.0.0.1

Destination = 8.8.8.8

Returning packet

8.8.8.8

↓

100.0.0.1

↓

R1

↓

192.168.0.167

R1 translates the destination back to the private address.

14. Static NAT — Key Characteristics

Advantages

- Allows private hosts to communicate over the Internet.
- Provides a permanent one-to-one mapping.
- Useful when an internal device needs a consistent public IP.
- Useful for servers that must be reachable using a predictable public address.

Disadvantage

It requires **one public IP for each private IP**.

Therefore:

Static NAT does not significantly conserve public IPv4 addresses.

15. CCNA Quick Revision

Private IPv4 ranges

10.0.0.0/8

172.16.0.0/12

192.168.0.0/16

NAT

NAT = Network Address Translation

Modifies source and/or destination IP addresses.

Static NAT

1 Private IP ↔ 1 Public IP

Inside Local

Private IP of inside host

Inside Global

Public/NAT IP of inside host

Outside Local

Outside host's address from local perspective

Outside Global

Outside host's address from outside perspective

Interface commands

ip nat inside

ip nat outside

Static NAT command

ip nat inside source static <inside-local> <inside-global>

Verification

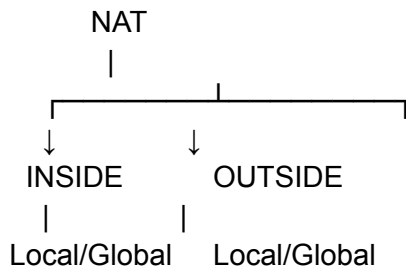
show ip nat translations

show ip nat statistics

Clear translations

clear ip nat translation *

★ Most Important Memory Trick



Inside/Outside = WHERE is the host?

Local/Global = FROM WHOSE PERSPECTIVE?

And for Static NAT:

INSIDE LOCAL		INSIDE GLOBAL
Private IP	↔	Public IP
192.168.0.167	↔	100.0.0.1
192.168.0.168	↔	100.0.0.2

One-line exam definition

NAT translates IP addresses between different address spaces, most commonly allowing private IPv4 hosts to communicate with the public Internet.

One-line Static NAT definition

Static NAT creates a permanent one-to-one mapping between an inside-local private IP and an inside-global public IP.

Dynamic NAT & PAT (NAT Overload)

1. NAT Basics

NAT (Network Address Translation) translates private/internal IP addresses into public/external IP addresses.

Important NAT terms

Term	Meaning
Inside Local	Private IP address of the internal host
Inside Global	Public IP representing the internal host
Outside Local	How the outside host appears to the inside network
Outside Global	Actual IP address of the outside host

Example

PC1: 192.168.0.167

↓ NAT

Public: 100.0.0.1

↓

Internet: 8.8.8.8

 **Dynamic NAT**

- In **dynamic NAT**, the router dynamically maps *inside local* addresses to *inside global* addresses as needed.
- An ACL is used to identify which traffic should be translated.
 - If the source IP is **permitted** by the ACL, the source IP will be translated.
 - If the source IP is **denied** by the ACL, the source IP will NOT be translated. *the traffic will NOT be dropped!
- A NAT pool is used to define the available *inside global* addresses that can be used.
- Although they are dynamically assigned, the mappings are still one-to-one (one *inside local* IP address per *inside global* IP address).
- If there aren't enough *inside global* IP addresses available (=all are currently being used), it is called 'NAT pool exhaustion'.
 - If a packet from another inside host arrives and needs NAT but there are no available addresses, the router will drop the packet.
 - The host will be unable to access outside networks until one of the *inside global* IP addresses becomes available.
 - Dynamic NAT entries will time out automatically if not used, or you can clear them manually.



2. Dynamic NAT

In **Dynamic NAT**, the router dynamically maps:

Inside Local → Inside Global

using a **NAT pool**.

Unlike Static NAT, the mapping is not permanently configured.

Three things are required

1. **ACL** → identifies which inside traffic should be translated.
2. **NAT Pool** → contains available public/inside-global addresses.
3. **NAT mapping command** → connects the ACL to the pool.

Example

ACL:

192.168.0.0/24

NAT Pool:

100.0.0.1 – 100.0.0.10

If:

192.168.0.167

is permitted by the ACL, the router may assign:

100.0.0.1

3. Important ACL Point

A very important exam point:

The NAT ACL identifies traffic to be translated. It does NOT act like a normal traffic-filtering ACL in this situation.

If the source IP is:

permit → translated

If the source IP is not permitted:

not translated

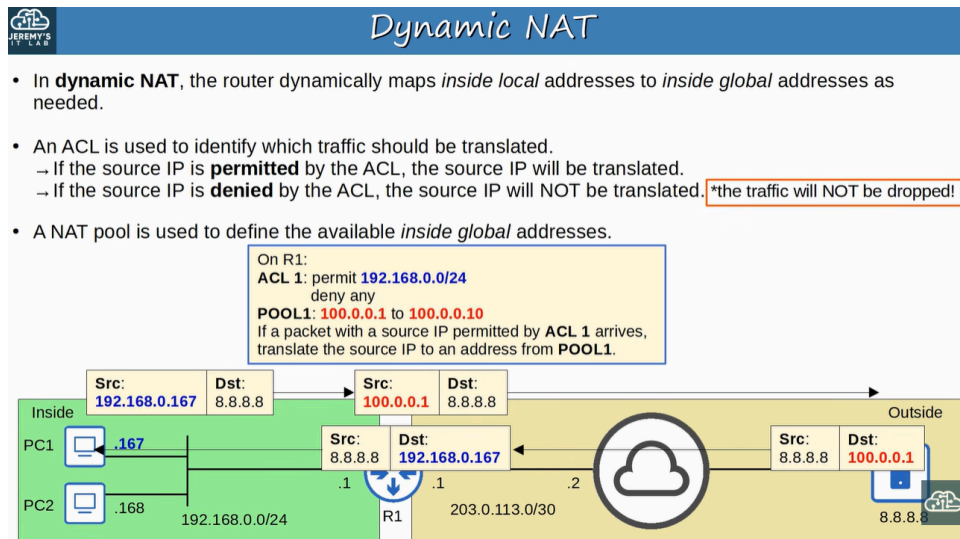
It does **not automatically mean the packet is dropped**.

Remember

NAT ACL = "Should I translate this?"

not necessarily:

"Should I allow this packet?"



4. NAT Pool

A **NAT pool** defines the available Inside Global addresses.

Example:

100.0.0.1 → 100.0.0.10

The router dynamically allocates these addresses to inside hosts.

Example:

192.168.0.168 → 100.0.0.1

192.168.0.100 → 100.0.0.2

192.168.0.12 → 100.0.0.3

192.168.0.28 → 100.0.0.4

Dynamic NAT normally provides a:

one-to-one mapping

1 Private IP ↔ 1 Public IP

5. NAT Pool Exhaustion

NAT Pool Exhaustion occurs when all public addresses in the NAT pool are already being used.

Example:

Pool = 100.0.0.1 – 100.0.0.10

There are only **10 available addresses**.

If all 10 are currently allocated and another internal host needs NAT:

192.168.0.98

there is no available Inside Global address.

Therefore:

No address available



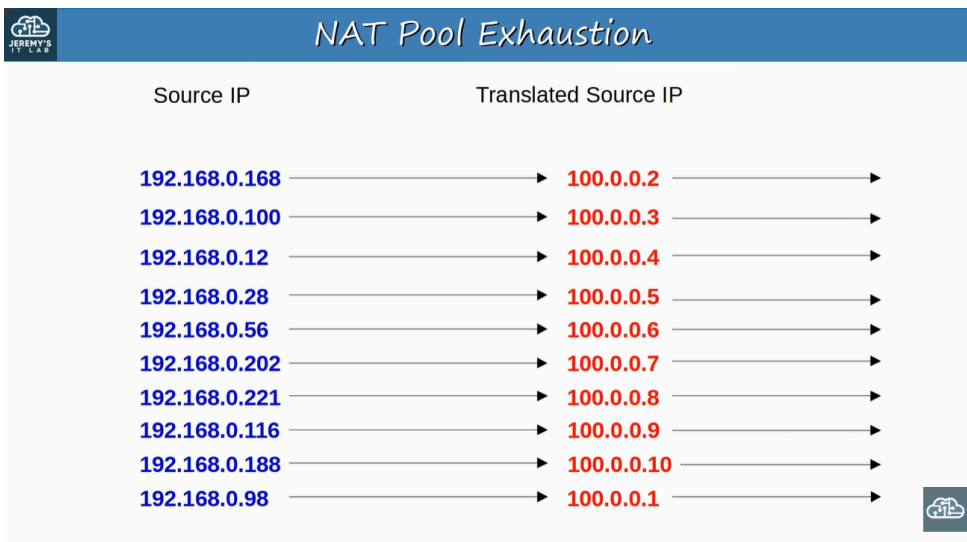
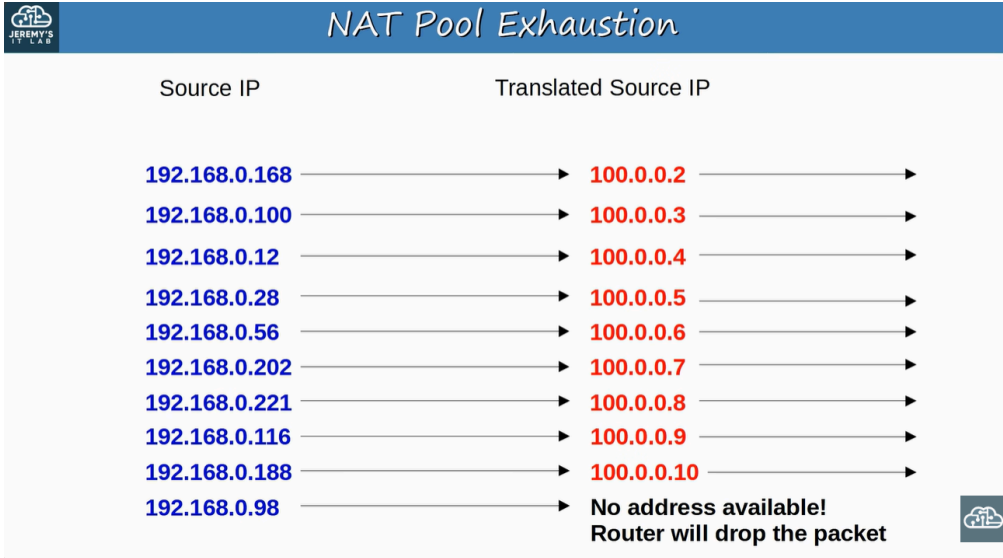
Router drops the packet

The host cannot access the outside network until an address becomes available.

Dynamic NAT entries

Unused dynamic NAT translations can:

- **Time out automatically**
- **Be cleared manually**



6. Dynamic NAT Configuration

Step 1 — Define Inside Interface

```
R1(config)# interface g0/1  
R1(config-if)# ip nat inside
```

This interface connects to the **internal/private network**.

Step 2 — Define Outside Interface

```
R1(config)# interface g0/0
R1(config-if)# ip nat outside
```

This interface connects to the **external/public network**.

Step 3 — Create ACL

```
R1(config)# access-list 1 permit 192.168.0.0 0.0.0.255
```

This identifies the inside addresses that should be translated.

Step 4 — Create NAT Pool

```
R1(config)# ip nat pool POOL1 100.0.0.0 100.0.0.255 prefix-length 24
```

or:

```
R1(config)# ip nat pool POOL1 100.0.0.0 100.0.0.255 netmask 255.255.255.0
```

Step 5 — Connect ACL to NAT Pool

```
R1(config)# ip nat inside source list 1 pool POOL1
```

Complete Dynamic NAT configuration

```
interface g0/1
ip nat inside
```

```
interface g0/0
ip nat outside
```

```
access-list 1 permit 192.168.0.0 0.0.0.255
```

```
ip nat pool POOL1 100.0.0.0 100.0.0.255 prefix-length 24
```

```
ip nat inside source list 1 pool POOL1
```



Dynamic NAT Configuration

```
R1(config)#int g0/1
R1(config-if)#ip nat inside

R1(config-if)#int g0/0
R1(config-if)#ip nat outside
R1(config-if)#exit

R1(config)#access-list 1 permit 192.168.0.0 0.0.0.255

R1(config)#ip nat pool POOL1 100.0.0.0 100.0.0.255 prefix-length 24

R1(config)#ip nat inside source list 1 pool POOL1
```

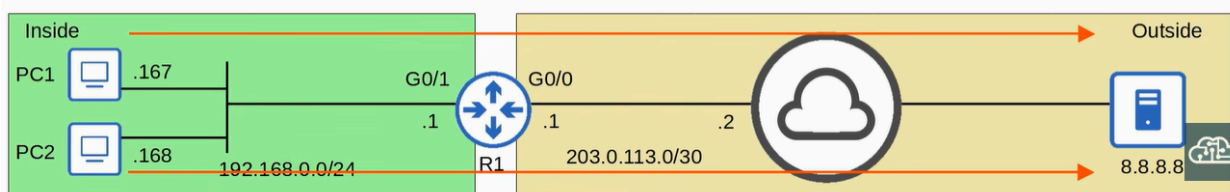
Define the 'inside' interface(s) connected to the internal network.

Define the 'outside' interface(s) connected to the external network.

Define the traffic that should be translated.
*Traffic permitted by this ACL will be translated.

Define the pool of inside global IP addresses.
*instead of **prefix-length 24**, you can use **netmask 255.255.255.0**

Configure dynamic NAT by mapping the ACL to the pool.



7. Verify Dynamic NAT

Show translations

R1# show ip nat translations

Example:

Pro	Inside global	Inside local
icmp	100.0.0.1	192.168.0.167
icmp	100.0.0.2	192.168.0.168

Meaning:

192.168.0.167 → 100.0.0.1
192.168.0.168 → 100.0.0.2

Show NAT statistics

R1# show ip nat statistics

This shows:

- Active translations
- Peak translations
- Inside interfaces
- Outside interfaces
- Hits
- Misses
- Dynamic mappings
- Pool utilization

Dynamic NAT Configuration

```

R1#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
icmp 100.0.0.1:3       192.168.0.167:3   8.8.8.8:3          8.8.8.8:3
udp 100.0.0.1:58685    192.168.0.167:58685 8.8.8.8:53         8.8.8.8:53
--- 100.0.0.1          192.168.0.167    ---               ---
icmp 100.0.0.2:3       192.168.0.168:3   8.8.8.8:3          8.8.8.8:3
udp 100.0.0.2:49536    192.168.0.168:49536 8.8.8.8:53         8.8.8.8:53
--- 100.0.0.2          192.168.0.168    ---               ---
  
```

Dynamic NAT Configuration

```

R1#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
icmp 100.0.0.1:3       192.168.0.167:3   8.8.8.8:3          8.8.8.8:3
udp 100.0.0.1:58685    192.168.0.167:58685 8.8.8.8:53         8.8.8.8:53
--- 100.0.0.1          192.168.0.167    ---               ---
icmp 100.0.0.2:3       192.168.0.168:3   8.8.8.8:3          8.8.8.8:3
udp 100.0.0.2:49536    192.168.0.168:49536 8.8.8.8:53         8.8.8.8:53
--- 100.0.0.2          192.168.0.168    ---               ---

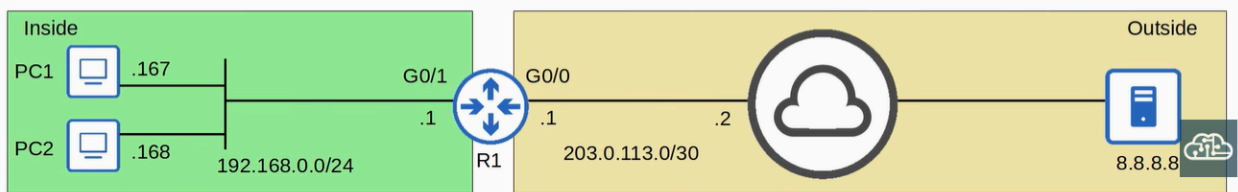
! below is about 1 minute later

R1#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
--- 100.0.0.1          192.168.0.167    ---               ---
--- 100.0.0.2          192.168.0.168    ---               ---
  
```



Dynamic NAT Configuration

```
R1#show ip nat statistics
Total active translations: 6 (0 static, 6 dynamic; 4 extended)
Peak translations: 6, occurred 00:00:30 ago
Outside interfaces:
  GigabitEthernet0/0
Inside interfaces:
  GigabitEthernet0/1
Hits: 32 Misses: 0
CEF Translated packets: 20, CEF Punted packets: 12
Expired translations: 0
Dynamic mappings:
-- Inside Source
[Id: 1] access-list 1 pool POOL1 refcount 6
  pool POOL1: netmask 255.255.255.0
    start 100.0.0.0 end 100.0.0.255
    type generic, total addresses 256, allocated 2 (0%), misses 0
[output omitted]
```



8. Why Dynamic NAT Entries Disappear?

Suppose:

192.168.0.167 → 100.0.0.1

192.168.0.168 → 100.0.0.2

Initially, active traffic creates detailed entries.

After the communication stops and the timeout occurs, the detailed translation disappears.

You may still see:

--- 100.0.0.1 192.168.0.167

--- 100.0.0.2 192.168.0.168

This represents the dynamic mapping information, but there is no currently active communication flow.

Key idea

Traffic active → detailed translation exists

Traffic stops → entry eventually expires

Image 7 — Paste here: Screenshot showing the NAT table before and after approximately one minute.

9. PAT — Port Address Translation

PAT is also called:

- NAT Overload
- NAT with overload

PAT translates:

IP address + Port number

This allows **many private hosts to share one public IP address**.

Example

PC1:

192.168.0.167:54321

PC2:

192.168.0.168:54321

Both can use the same public IP:

100.0.0.1

but PAT assigns different port numbers:

192.168.0.167:54321

↓

100.0.0.1:54321

192.168.0.168:54321

↓

100.0.0.1:54322

Therefore the router can distinguish the two sessions.

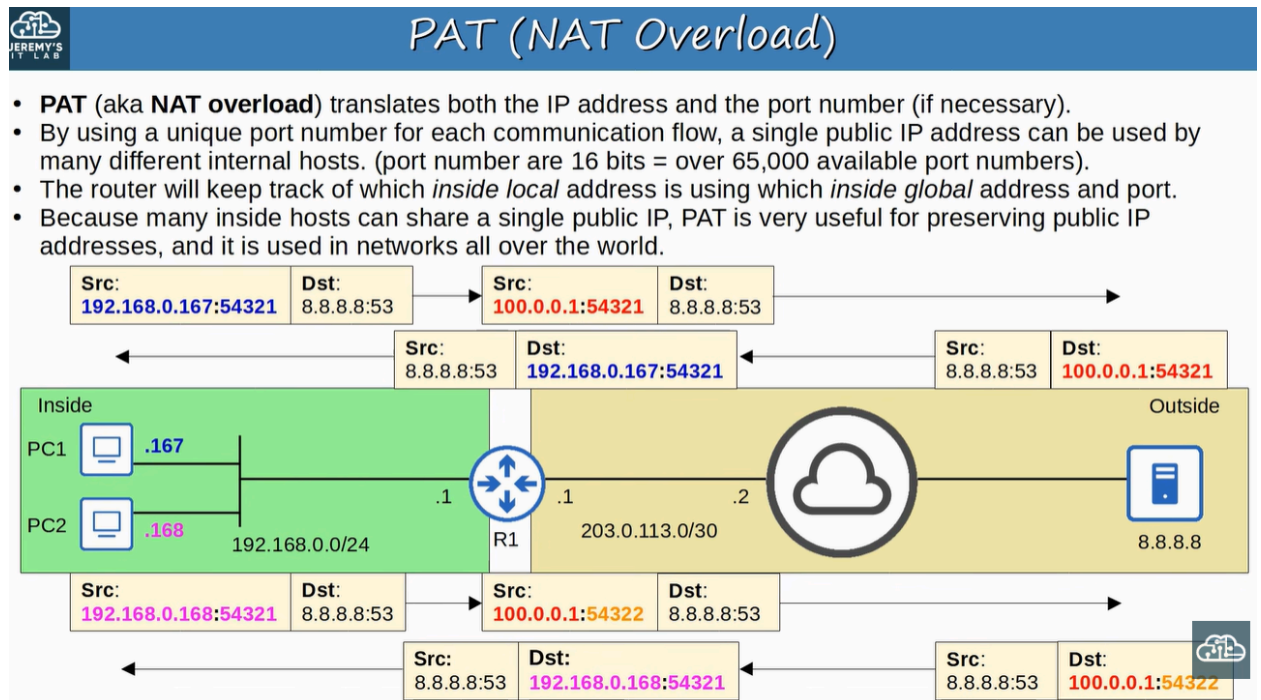
Core idea ★

Dynamic NAT:

1 private IP ↔ 1 public IP

PAT:

Many private IPs ↔ 1 public IP
using different ports



10. PAT Using a NAT Pool

PAT can use a pool of public addresses.

Configuration

```
interface g0/1  
ip nat inside
```

```
interface g0/0  
ip nat outside
```

```
access-list 1 permit 192.168.0.0 0.0.0.255
```

```
ip nat pool POOL1 100.0.0.0 100.0.0.3 prefix-length 24
```

ip nat inside source list 1 pool POOL1 overload

Important

The keyword:

overload

makes it **PAT**.

Without:

overload

it is **Dynamic NAT**.

Easy memory trick

POOL only

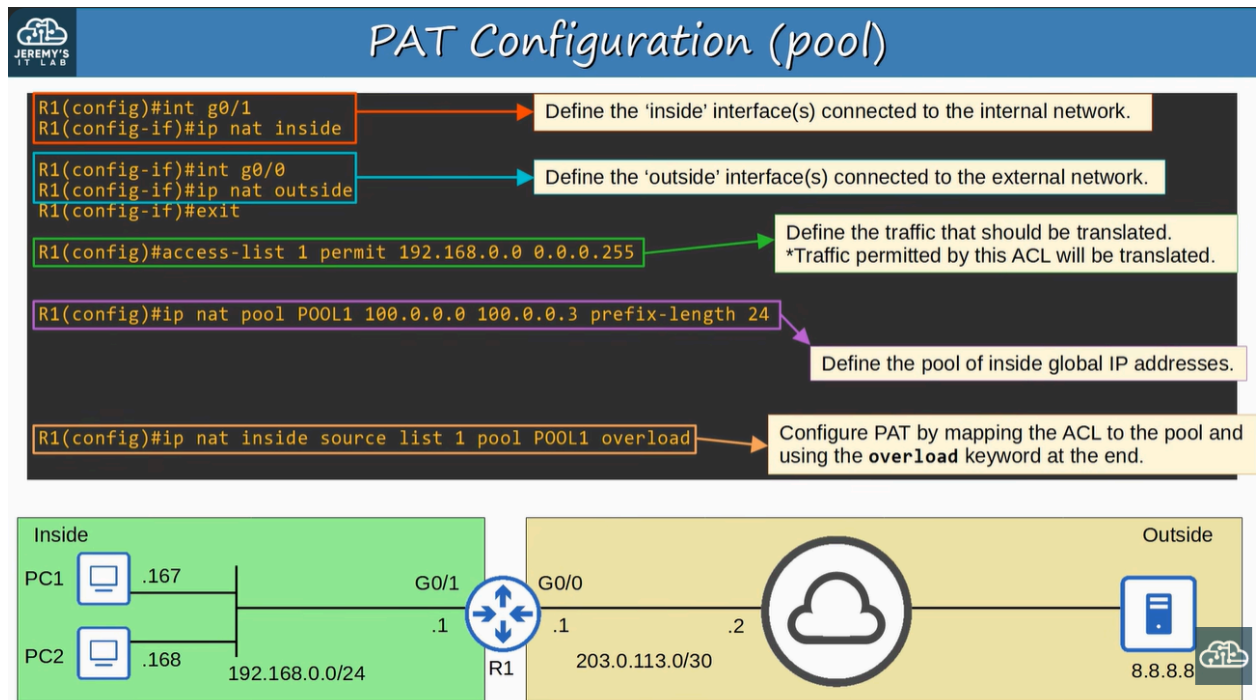
↓

Dynamic NAT

POOL + overload

↓

PAT



11. PAT Using the Outside Interface

Instead of creating a separate NAT pool, PAT can use the router's outside interface IP.

```
R1(config)# ip nat inside source list 1 interface g0/0 overload
```

This means:

Translate all permitted inside hosts using the IP address of `GigabitEthernet0/0`, and distinguish their connections using port numbers.

Example

Outside interface:

203.0.113.1

PC1:

192.168.0.167:65205

becomes:

203.0.113.1:65205

PC2:

192.168.0.168:59641

becomes:

203.0.113.1:59641

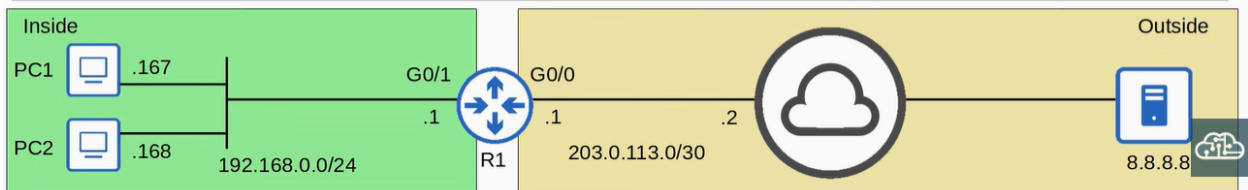
Both hosts are sharing:

203.0.113.1



PAT Configuration (pool)

```
R1#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
udp 100.0.0.1:63925    192.168.0.167:63925 8.8.8.8:53         8.8.8.8:53
udp 100.0.0.1:59549    192.168.0.168:59549 8.8.8.8:53         8.8.8.8:53
R1#show ip nat statistics
Total active translations: 2 (0 static, 2 dynamic; 2 extended)
Peak translations: 2, occurred 00:00:03 ago
Outside interfaces:
  GigabitEthernet0/0
Inside interfaces:
  GigabitEthernet0/1
Hits: 4 Misses: 0
CEF Translated packets: 0, CEF Punted packets: 4
Expired translations: 0
Dynamic mappings:
-- Inside Source
[Id: 3] access-list 1 pool P00L1 refcount 2
  pool P00L1: netmask 255.255.255.0
    start 100.0.0.0 end 100.0.0.3
    type generic, total addresses 4, allocated 1 (25%), misses 0
```



PAT Configuration (interface)

```
R1(config)#int g0/1
R1(config-if)#ip nat inside
```

Define the 'inside' interface(s) connected to the internal network.

```
R1(config-if)#int g0/0
R1(config-if)#ip nat outside
R1(config-if)#exit
```

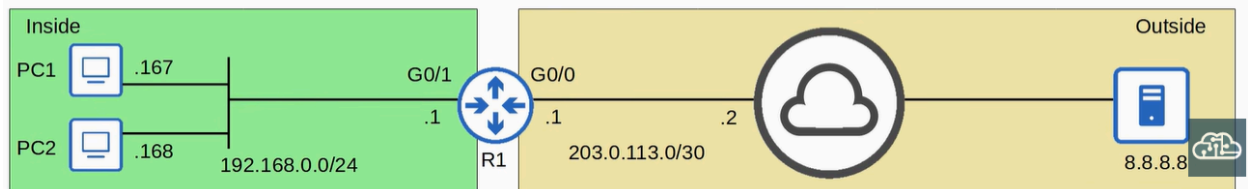
Define the 'outside' interface(s) connected to the external network.

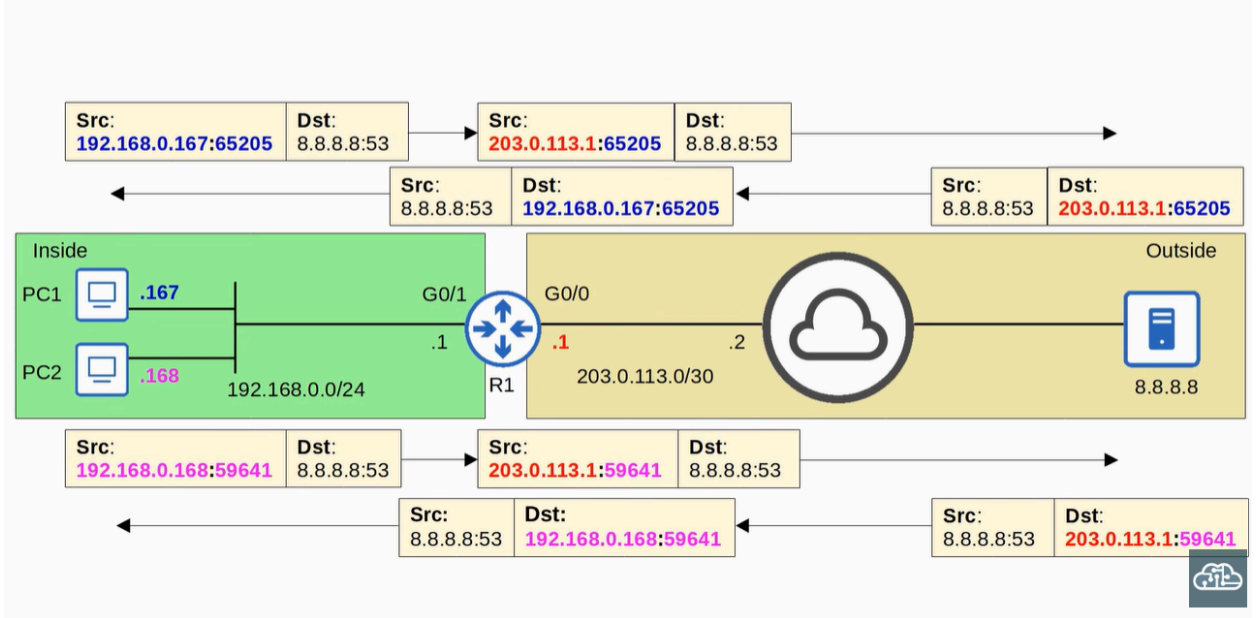
```
R1(config)#access-list 1 permit 192.168.0.0 0.0.0.255
```

Define the traffic that should be translated.
*Traffic permitted by this ACL will be translated.

```
R1(config)#ip nat inside source list 1 interface g0/0 overload
```

Configure PAT by mapping the ACL to the interface and enabling **overload**.





12. PAT Translation Table

Use:

R1# show ip nat translations

Example:

```
Pro Inside global    Inside local
udp 203.0.113.1:65205 192.168.0.167:65205
udp 203.0.113.1:59641 192.168.0.168:59641
```

The router remembers:

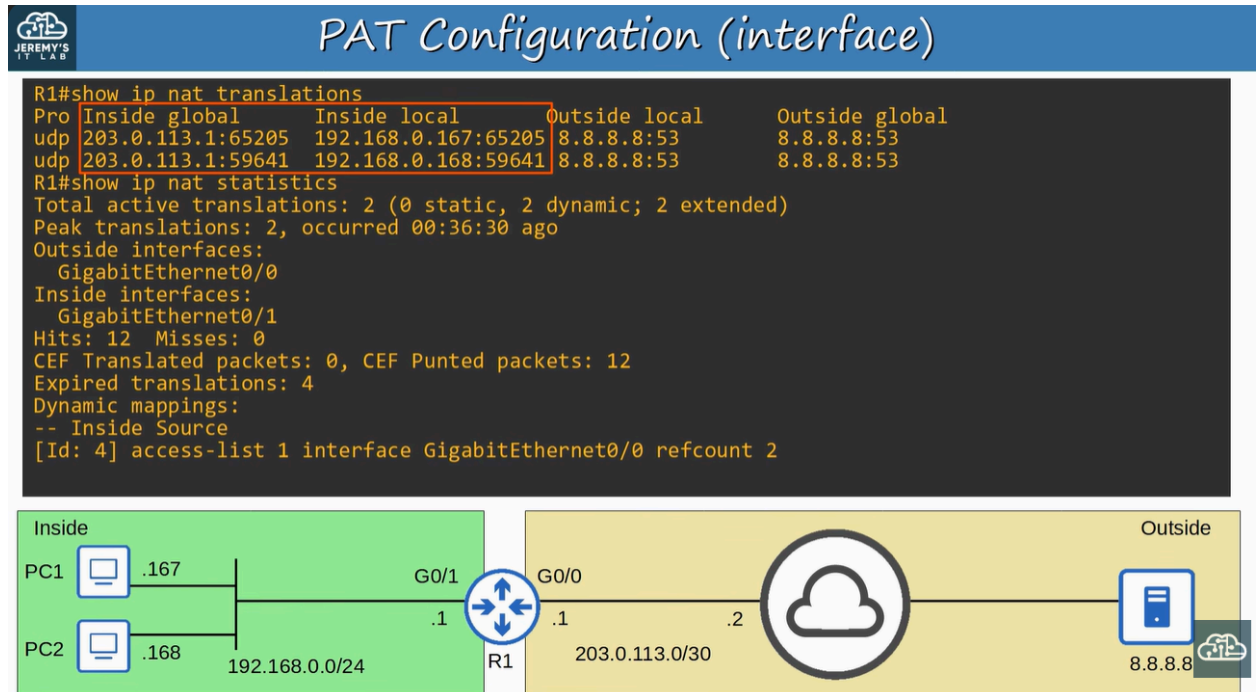
Public IP + Port
 ↑
 Private IP + Port

When the reply comes back:

Internet
 ↓
 203.0.113.1:65205

↓
192.168.0.167:65205

The router knows exactly which internal host should receive the packet.



13. Dynamic NAT vs PAT ★★

Feature	Dynamic NAT	PAT
Translation	IP address	IP + Port
Mapping	One-to-one	Many-to-one possible
Public IP requirement	More public IPs	Can use one public IP
Keyword	No overload	overload
Example	192.168.0.167 → 100.0.0.1	192.168.0.167:5000 → 203.0.113.1:5000
Main purpose	Dynamic public IP allocation	Share public IP among many hosts

Dynamic NAT

ip nat inside
ip nat outside

access-list 1 permit 192.168.0.0 0.0.0.255

ip nat pool POOL1 100.0.0.0 100.0.0.255 prefix-length 24

ip nat inside source list 1 pool POOL1

PAT using pool

ip nat inside source list 1 pool POOL1 overload

PAT using interface

ip nat inside source list 1 interface g0/0 overload

Verification

show ip nat translations
show ip nat statistics

Clear translations

clear ip nat translation *

16. Important Doubts / Clarifications

Doubt 1: Does an ACL deny drop the packet?

Not necessarily.

For a NAT ACL:

Permit → translate

Deny/not matched → don't translate

The NAT ACL is being used to determine **which traffic gets NATed**.

Doubt 2: Why does Dynamic NAT need a pool?

Because Dynamic NAT needs a set of **Inside Global/public addresses** from which the router can dynamically allocate an address.

Private IP
↓
NAT Pool
↓
Available public IP

Doubt 3: Why does PAT not need a large public pool?

Because PAT uses **port numbers** to distinguish different connections.

Therefore:

Many private hosts
↓
One public IP
+
Different port numbers

Doubt 4: What happens when a Dynamic NAT pool is exhausted?

If every public address is currently allocated:

New inside host
↓
No public IP available
↓
Packet is dropped

The host must wait until a NAT address becomes available.

Doubt 5: Why can two hosts use the same public IP in PAT?

Because their **port numbers are different**.

Example:

192.168.0.167:54321

↓
100.0.0.1:54321

192.168.0.168:54321

↓
100.0.0.1:54322

The router uses the IP + port combination to distinguish the sessions.

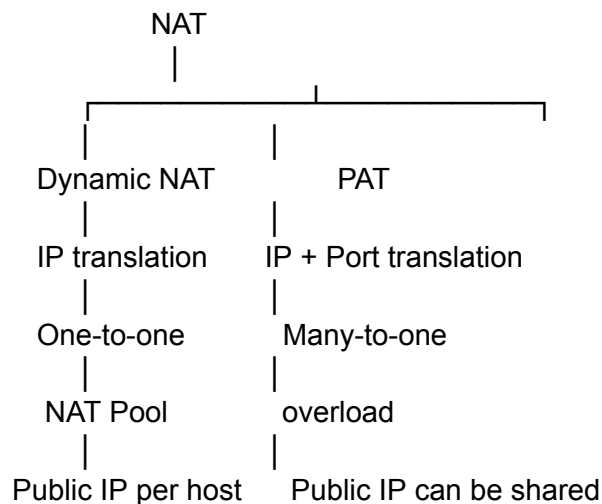
Doubt 6: What does **overload** mean?

overload = PAT

It tells the router:

Allow multiple inside hosts to share the same public IP by using different port numbers.

17. Final Memory Map ★



One-line exam definitions

Dynamic NAT:

Dynamically maps inside local addresses to available inside global addresses from a NAT pool.

PAT:

Allows multiple inside hosts to share one or more public IP addresses by using different port numbers.

NAT Pool Exhaustion:

Occurs when all available inside global addresses in the NAT pool are currently allocated.

